Software Requirements Specification

for

JeRango Warehouse Manager

Version 2.0.0

Prepared by

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1 Introduction

This document will encompass the project entitled **JeRango Warehouse Manager,** an Inventory Software Management System [ISMS] responsible for tracking dry goods and products of the warehouse. This section provides a detailed description of the document, from its purpose and scope to an overview of the document and the terminologies used.

1.1 Document Purpose

The purpose of the software requirements specification [SRS] document is to provide the readers with a background and knowledge of the ISMS, particularly on the specific requirements the software needs. This document will expound on the varying features of the system, specifically its purpose on the overall functionality of the software. The limitations and constraints of the system are to be enumerated and explained thoroughly.

These detailed descriptions of the features and limitations on both the user interface [UI] and user experience [UX] sides may provide a comprehensive and understandable view of the varying requirements of the system.

1.2 Product Scope

The ISMS catalogs the incoming and outgoing goods. The main proponent of the system is a database capable of handling large quantities of data and information on inventory content. Through this, the users of this software can efficiently organize their inventory and ensure transparency.

Aside from the system's primary feature, the system notifies users of expired items, items of low stock, and the number of items currently available. The system should also track incoming and outgoing orders. Excess stocks are stored in a separate inventory and will be used as replenishment before new stocks arrive.

1.3 Intended Audience and Document Overview

This SRS document presents the product and its overall description. The second chapter provides further exposition regarding this matter. Chapter three delves into a comprehensive look into the requirements, features, and functionality of the developed software. The fourth section thoroughly lists and defines the non-functional requirements of the software.

The intended audiences of this SRS are the following: the clients and the professor. Separated into subsections are the developers' suggestions as to the sequence and flow of how each type of reader should begin reading the document.

1.3.1. The Clients

The clients should immediately jump to the product overview section of the document as it provides a detailed summary of the ISMS. It is followed by reading through the overview of the software's functionality and cross-referencing it to its corresponding section in the third chapter. They may opt for reading the design constraints section of the document, as these features in the third chapter will only be implicitly stated or reiterated.

1.3.2. The Professor

The professor would have to read the entire document, top to bottom, to ensure its credibility, cohesiveness, and plausibility. By reading the entire document, the authors mean to go through the second chapter first, to confirm that only high-level summaries are written under its subsections. The third and fourth chapters of this document must be detailed and well-written, as it contains the most essential parts of the document.

In addition, the authors suggest, to all readers, skimming through the Definitions, Acronyms, and Abbreviations section whenever needed. The listed terms will help the readers understand the full context of certain phrases and sentences in each of the sections.

1.4 Definitions, Acronyms, and Abbreviations

To easily and fully comprehend this software requirements specification document, the terminologies, acronyms, and abbreviations are categorized and listed below.

1.4.1 Terminologies and their Definition.

Item Catalog. A catalog of goods, property, and merchandise held in stock. May also be referred to as a product database.
Manager. The intended user of the ISMS.
Management system. The method by which an organization oversees the interconnected components of its operations to attain its objectives.
Warehouse. A large building for storing things before they are sold, used, or sent out to stores.

1.4.2 Acronyms.

GUI. Graphical user interface.
IEEE. Institute of Electrical and Electronics Engineers
ISMS. Inventory Software Management System
SRS. Software Requirements Specification
UI. User Interface
UX. User Experience

1.5 Document Conventions

This SRS follows the IEEE formatting requirements throughout the document. The developers utilized the Arial font throughout the document, with varying font sizes to determine hierarchy. The main headline uses a font size of 18. The subcategory titles and the contents of each section have a font size between 12 to 14 and 11, respectively. It maintains a 1” margin on all sides, and the text is single-spaced.

Acronyms, when first encountered, have their full name spelled out. Its full meaning is accompanied by its appropriate acronym enclosed in brackets.

1.6 References and Acknowledgments

[1] W. Kenton, “Inventory,” Investopedia, May 27, 2022. https://www.investopedia.com/terms/i/inventory.asp

[2] ISO, “Management system standards,” ISO, 2011. https://www.iso.org/management-system-standards.html

[3] Cambridge Dictionary, “warehouse,” @CambridgeWords, Jan. 25, 2023. https://dictionary.cambridge.org/us/dictionary/english/warehouse

[4] “FREE IEEE Citation Generator [Updated for 2020],” MyBib. https://www.mybib.com/tools/ieee-citation-generator

[5] https://www.cse.chalmers.se/~feldt/courses/reqeng/examples/srs_example_2010_group2.pdf

[6] https://www.se.rit.edu/~co-operators/Semester2/SoftwareRequirementsSpecification.pdf

2 Overall Description

Under this section is a description of the inventory software management system. It details an in-depth overview of the product and its purpose. A summary of all the main functions of the software is listed here. Additionally, a list of limitations, assumptions, and dependencies is added in this section to inform the readers of the limiting factors that may or will affect the development of the product.

2.1 Product Overview

JeRango Warehouse Manager is an ISMS self-contained product with a database of goods and products. The database allows the administrator to view, filter out, catalog, restock, and remove products, or simply track the status of the goods. Another major feature is the order log, allowing the user to view the history of orders initially filtered based on recency. The figure below shows an overview of the interactions between the user, the system, and the subsystems:

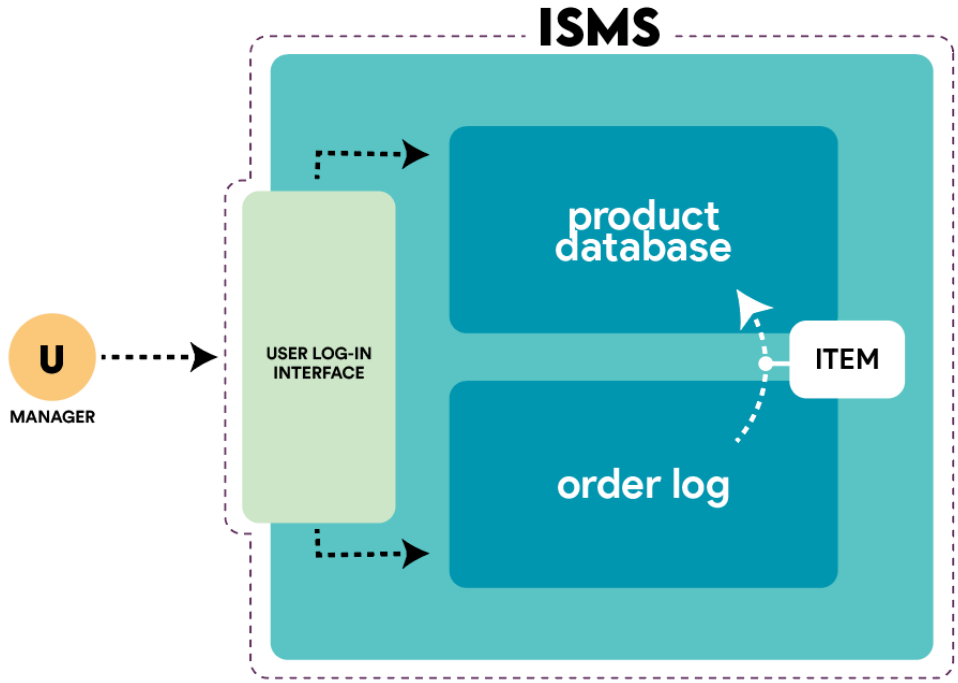

Figure 1. System Environment of the ISMS

The ISMS has one active actor (manager), and one software system (the ISMS) that is composed of several subfunctions. The manager must successfully pass through the user log-in interface before he/she can access the ISMS. The manager can access the two main features of the system: the product database and the order log. In the product database, the manager can track the status of the products. On the other hand, the manager can view the history of recent orders in the order log.

One additional major interaction is between the product database and the order log. Once an order is recorded in the order log, its list of items will be added to the product database.

2.2 Product Functionality

The following list showcases the main functions the product must perform or let the user perform when using the ISMS. These functionalities will be expounded in the third chapter of this SRS document.

1. **User Authentication.** It ensures that only authorized users can access the system.
2. **Item Catalog.** It displays a list of items that can be filtered based on name, supplier, expiration date, price, and serial number.
3. **Order Management.** It showcases a list of orders based on recency, including information such as the supplier's name, quantity, date of request, and date of arrival.
4. **Inventory Analysis and Reporting.** A notification system that reports the user of to-be-expired products and products of low inventory, and displays the item quantity.
5. **Data Backup and Recovery.** It recovers all data and information in a backup file to ensure no loss of information.

2.3 Design and Implementation Constraints

Generally, JeRango Warehouse Manager is designed with small-scale businesses in mind but not limited to it. Although it can handle the data and information on the scale of a warehouse, its current state is not fitting for interstate and global businesses and corporations. Aside from the general constraints mentioned above, below is a list of all constraints, sorted into categories, that the developers are limited to.

2.3.1 Hardware Constraints

These are the restrictions imposed by the components of a computer system or electronic device that can impact the performance and capabilities of the ISMS. The developers identified the following hardware constraint:

1. The ISMS doesn’t utilize hardware parallel operation for other functionalities like automatic inventory tracking and hardware-supported expiry date counters.
2. Limit testing may prove limited due to the available resources.
3. The ISMS is built on the Windows 10, 64-bit, 4GB RAM, CPU’s 2014 and beyond with no plans to expand cross-platform.

2.3.2 Design Constraints

Design constraints are the internal and external restrictions imposed in the design process. These limitations will allow the developers to narrow their scope of ideas and help them develop a feasible design. The project-specific design constraints are the following:

1. The database of the ISMS is designed to accommodate a ‘city-wide’ scale.
2. The application is specialized to the clients’ industry of holding.
3. The application will incorporate a neomorphic design convention to conform to modern standards.

4. Google's Firebase holds a database and is incorporated into JavaScript, CSS, and HTML to create the UX and UI of the application.

2.3.3 Implementation Constraints

Implementation constraints are imposed restrictions on the implementation process of the ISMS. The following are the identified implementation constraints by the developers:

1. Third-party software utilization may only be optimized for select software.
2. Maintenance is passed over to the client after submission.
3. Moderate work will be done to the security of the application

2.4 Assumptions and Dependencies

Assumptions and Dependencies allow the developers to lay the foundations of the design and implementation of the project. The following list displays the factors that may significantly affect the ongoing processes of the JeRango Warehouse Manager. It is to be noted that the project could be affected if these assumptions and dependencies aren't shared.

2.4.1 Assumptions

Project assumptions are the expectations and predictions that the developers have identified while conceptualizing the project. Here is a list of the identified assumptions:

1. The maximum scale of the ISMS is undefined due to unmaximized limit testing.
2. The ISMS may not be a potential substitute for similar use case utilizations such as an establishment population manager, mass transport capacity manager, and parking space system managers.
3. It may not be efficient in businesses that are drastically different from the client's business.

2.4.2 Dependencies

Project dependencies, internal or external, are the relationships between the different system functions based on their sequence. They provide a clear picture of how the different subsystems are related.

1. Firebase Database was utilized in recording the data.
2. System Clock is utilized for the tracking of stock expiry date
3. System Notification Manager is required for the expiry date notification built into the application
4. A basic functioning back-end side of the system must be initially developed before integrating it with the front-end side of the system.

3 Specific Requirements

Under this section are the specific requirements needed for the ISMS to completely function. Listed here are the external interface requirements, sorted into two categories: user interfaces and software interfaces. The functional requirements as listed in section 2.2 will be expounded in detail in this section as well. Use case models are generated to identify a framework and create a flow of the functionalities of the system. Despite this being a software application

3.1 External Interface Requirements

External Interface requirements include the following interfaces specific to the needs of the ISMS: user interfaces and software interfaces. User interfaces refer to the interaction logic between the software and the user. Software interfaces refer to the software applications the developers are utilizing to communicate with each other.

As mentioned in the previous section, the developers will not be integrating any hardware components for automatic inventory tracking and hardware-supported expiry date counters. Hence, hardware interfaces are not needed in this project.

3.1.1 User Interfaces

The JeRango Warehouse Manager shall utilize a graphical user interface [GUI] as it allows the user to interact with the graphical elements of the designed system. It is through this interface that the developers can create most of the main functions of the ISMS, mainly the item catalog and the order log. The following figures (Figure 2, 3, and 4) below will represent a framework of the major user interfaces to be developed in this project.

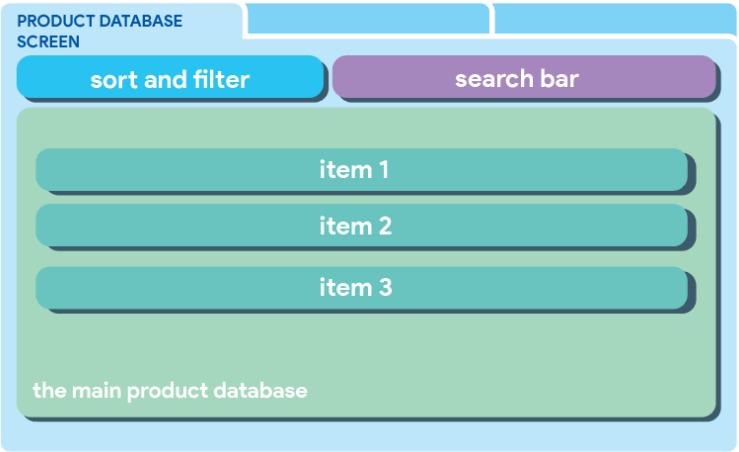

Figure 2. Framework User Interface for the Product Database Screen

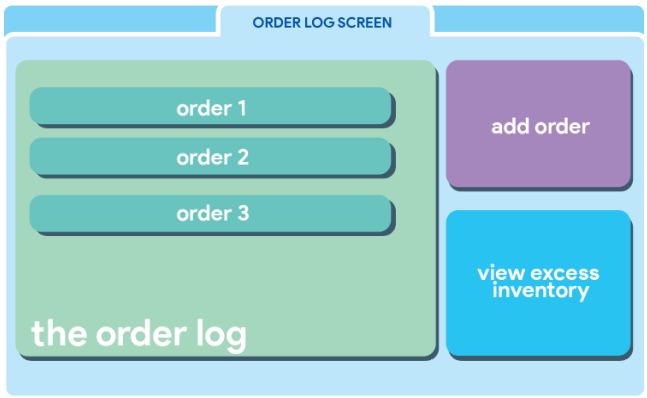

Figure 3. Framework User Interface for the Order Log Screen

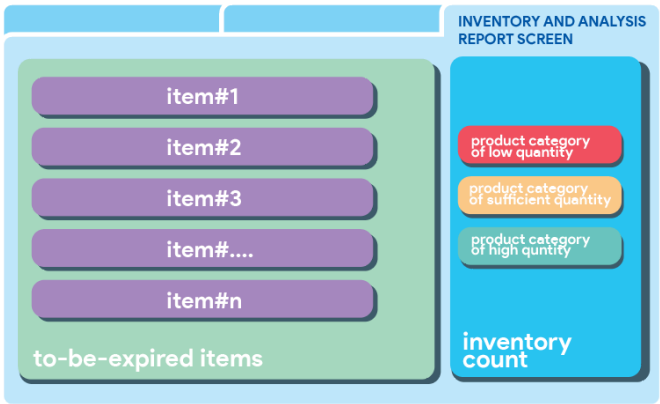

Figure 4. Framework User Interface for the Inventory Analysis and Report Screen

3.1.2 Software Interfaces

JeRango Warehouse Manager will need an operating system and a database. Hence, an Application Programming Interface [API] is utilized. For the back-end side of the application, SQL and Java programming languages are used. In the front end, the web design mechanics of HTML and CSS are useful in creating a neomorphic design. PHP is utilized in saving the data from HTML to SQL.

3.2 Functional Requirements

The following functional requirements capture the intended behavior of the ISMS. Under this section are the in-depth descriptions of the main functions of the system as listed in section 2.2. Specifically, this section will delve into the following functionalities: the User Authentication System, the Item Catalog, the Order Log, the Inventory Analysis and Report System, and the Data Recovery System.

3.2.1 The User Authentication System

This system allows the software to identify which users are verified so that unauthorized users will be blocked from accessing sensitive information from the software. The user's information will be stored in a different file such as their username and password. With this system, the user's

information will be protected and only be accessed with their account and possibly accounts that have higher roles than a user such as a moderator.

3.2.2 The Item Catalog

The most crucial function of the software is the item catalog. Item cataloging allows the user to create items, add descriptive details about it, and systemically arrange them but each item that will be delivered should have a unique identifier such as a serial number. The system can also allow users to search for a certain item.

3.2.3 The Order Log

The order log is a report that lets the user view supplier orders. Since this is a tracking system, it will display the supplier name, type and quantity of items ordered, date of request, and date of arrival. The status of the item should also be reported if it is out of stock so that suppliers can be notified early and restock the items or when excess items will be used to restock the items.

3.2.4 The Inventory Analysis and Report System

Inventories must be regularly audited so that it will be ensured that the process is running smoothly. With this system, it can aid the user in maintaining the right amount of stock to keep on-hand to successfully fill in the demands created by customers, all while avoiding to overly invest on inventory storage. The system also allows users to save a ton of money since it cuts costs and reduces the risk of running out of stock. Basically, the functionality of this system lists down the stock's status which are its expiration date, and quantity.

3.2.5 The Data Recovery System

As the name suggests, this system creates a backup that stores all necessary data and information so that we will reduce the risk of losing all information. The data will be then restored from a backup file once the data has been lost, corrupted or made inaccessible.

3.3 Use Case Model

The developers have identified only one user described in the requirements of the clients. This user is the 'Manager'. Under this user are several use cases about the different subfunctions of the system. These are the identified manager use cases in the system.

Table 1. The Identified Use-Cases Under Each Main Functionality

Functions	User Authentication	Product Database	Order Log	Inventory Analysis and Report
Identified Use Cases	• User Authentication	• view product database • search product • sort products	• view order log • add order • edit order status • view excess inventory	• Pop-up Report • View Analysis Report

3.3.1 Manager Use Case - User Authentication [MUC - UA]

Figure 5. Diagram for Use Case MUC - UA.

Author/s: Cyril Andre Durango, Ivor Louisetyne Canque
Purpose: To authenticate the user. Protects the software and database from unauthorized access.

Requirements Traceability:

Priority: Higher Priority. It precedes every function of the application, acting as a barrier against wrongful access.

Preconditions: Opening the application
Postconditions: The Manager is immediately directed to the product database screen of the ISMS

Actors: Manager
Extends: N/A

Flow of Events:

1. Basic Flow
   1.1. The manager opens the app.
   1.2. Manager inputs required credentials.
   1.3. Software checks if credentials match.
   1.4. Once the credentials match, access to the application is granted.

2. Alternative Flow
   2.1. In step 4, if the inputted credentials do not match, the software allows five (5) chances for reauthentication.
   2.2. If the Manager fails to input the correct credentials and has used up all 5 chances of reauthentication, he or she will be unable to log in to the system for 15 minutes.
   2.3. After the 15 minutes are up, the Manager can log in once again.

3. Exceptions. The log-in attempt may be abandoned at any time.

Includes: MUC - IAR - PuR
Notes/Issues: N/A

3.3.2 Manager Use Case - Product Database - View Product Database [MUC - PD - VPD]

Figure 6. Diagram for Use Case MUC - PD - VD

Author/s: Cyril Andre Durango
Purpose: Allows the manager to view all products currently stored in the database

Requirements Traceability:

Priority: Highest Priority. This is the main function of the ISMS, thus its completion is crucial to the success of the application.

Preconditions: The Manager has authenticated access to the database, or the Manager is on another screen that is not the product database screen.
Postconditions: The Product Database Screen is displayed.

Actors: Manager
Extends: N/A

Flow of Events:

1. Basic Flow
   1.1. Once the Manager is granted access, they are immediately directed to the product database screen.
   1.2. On the upper portion of the database are the different tabs of the different functionalities of the ISMS, currently highlighted is the Product Database. The other tabs include the Order Log and the Inventory Analysis Tab.
   1.3. The Manager can scroll through all products of the database.
2. Alternative Flow
   2.1. In the case that the Manager is on a different screen, they can just choose the Product Database Tab to be redirected back to the Product Database screen.

3. Exceptions
   3.1. The Manager may exit the application at any time by logging out.
   3.2. The Manager may choose another tab to redirect themselves away from the Product Database screen.

Includes: N/A
Notes/Issues: N/A

3.3.3. Manager Use Case - Product Database - Search Product [MUC - PD - SP]

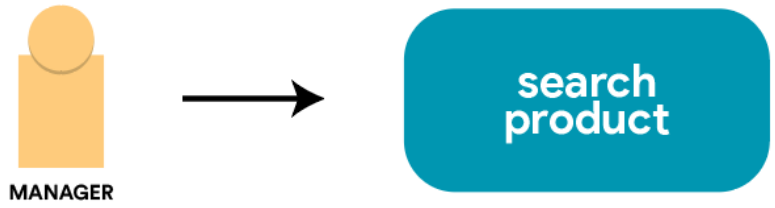

Figure 7. Diagram for Use Case MUC - PD - SP

Author/s: Cyril Andre Durango
Purpose: It allows the manager to search products based on serial number, name, supplier, price, and expiration date.

Requirements Traceability:

Priority: Higher Priority. Searching products allows for filtering which products the Manager wants to view.

Preconditions: The Manager has been authenticated access to the database
Postconditions: N/A

Actors: Manager
Extends: MUC - PD - VD

Flow of Events:

1. Basic Flow
   1.1. The Manager clicks on the search bar and searches for the serial number of a certain product
   1.2. Once the serial number is entered, the corresponding product must be displayed in the interface.

2. Alternative Flow
   2.1. In step 1, the manager may choose to search through the name of the product, the supplier, the price, and the expiration date.
   2.2. If the Manager opts to search for the name of the product, the database must display all products of that name.
   2.3. If the Manager opts to search for the supplier of that product, the database must display the products coming from that certain supplier.
   2.4. If the Manager opts to search for the price, the database must display the products of that price.
   2.5. If the Manager opts to search for an expiration date, the database must display the products of the indicated expiration date.

3. Exceptions. The Manager may abandon the search process at any time.

Includes: N/A
Notes/Issues: N/A

3.3.4. Manager Use Case - Product Database - Sort Products [MUC - PD - SoP]

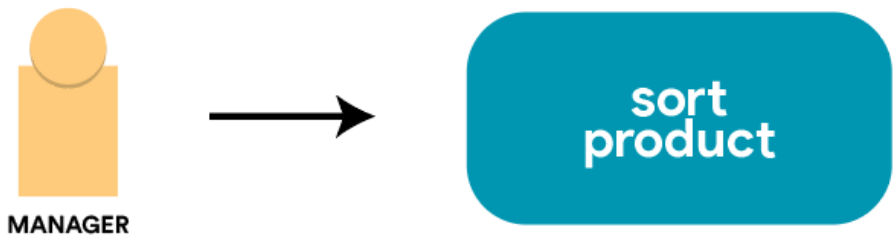

Figure 8. Diagram for Use Case MUC - PD - SoP

Author/s: Cyril Andre Durango
Purpose: Allows the manager to sort the products of the database by serial number, alphabetical order of product or supplier, price, and expiration date.

Requirements Traceability:

Priority: Medium Priority. Often executed, it isn't required by other use cases

Preconditions: The products in the database are sorted based on serial numbers and are in alphabetical order.
Postconditions: The products are sorted based on the filter chosen by the Manager

Actors: Manager
Extends: MUC - PD - VPD
MUC - PD - SP

Flow of Events:

1. Basic Flow
   1.1. The Manager can choose to sort the products of the database.
   1.2. The system sorts out the products based on the chosen filter.
   1.3. The Manager can see the rearranged products based on its filter.

2. Alternative Flow
   2.1. In the instance that the Manager has already searched under the following options as stated in the MUC-PD-SP use case, the system must also consider the search output when filtering.
   2.2. If the Manager chooses to sort the product name or supplier name alphabetically, it will display the products in alphabetical order or reverse alphabetical order based on the filter they chose.
   2.3. The Manager can opt to sort the products by their expiration date.
   2.4. The Manager can also choose to sort the products by price, from the lowest price to the highest price, and vice-versa.

3. Exceptions. The Manager can opt to not use the filter when viewing the database.

Includes: N/A
Notes/Issues: N/A

3.3.5. Manager Use Case - Order Log - View Order Log [MUC - OL - VOL]

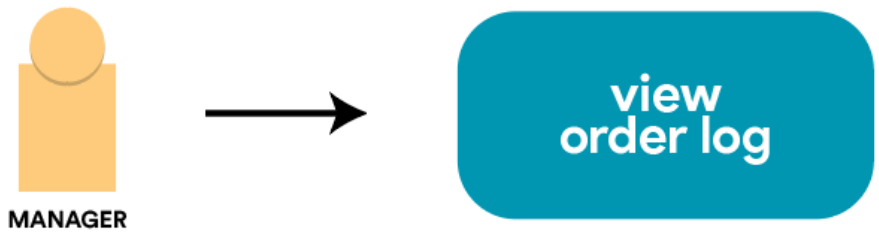

Figure 9. Diagram for Use Case MUC - OL - VOL

Author/s:
Purpose: Allows the manager to view all orders currently stored in the order log database

Requirements Traceability:

Priority: Higher Priority. This is the next main functionality of the system.

Preconditions: The Manager must be on another screen from the Order Log screen
Postconditions: The Order Log screen is displayed.

Actors: Manager
Extends: N/A

Flow of Events:

1. Basic Flow
   1.1. The Manager clicks the Order Log located on the upper portion of the screen
   1.2. The system displays the order log.

2. Exceptions
   2.1. The Manager may choose another tab to redirect themselves away from the Order Log screen
   2.2. The Manager may choose to exit and log out of the application.

Includes: N/A
Notes/Issues: N/A

3.3.6. Manager Use Case - Order Log - Add Order [MUC - OL - AO]

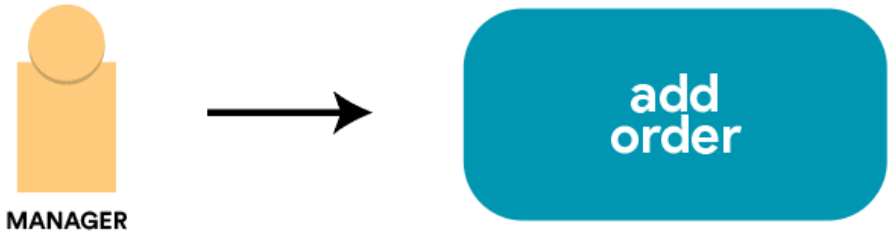

Figure 10. Diagram for Use Case MUC - OL - AO

Author/s:
Purpose: Allows the manager to add orders to be stored in the order log.

Requirements Traceability:

Priority: High Priority. Allows for transparency in the orders and the items being ordered.

Preconditions: The manager must be in the Order Log Screen.
Postconditions: A new order is added to the order log with a default status of "ordered".

Actors: Manager
Extends: MUC - OL - VOL

Flow of Events:

1. Basic Flow
   1.1. The Manager clicks a 'plus' button to add a new order.
   1.2. A new window will pop-up which detail a sel
   1.3. The Manager can add multiple product categories and their corresponding quantity to one order.
   1.4. The Manager must also encode the name of the supplier.
   1.5. The Manager confirms the ordered items by clicking another button.
   1.6. The system creates a new instance of an order in the Order Log, with its default status set to "ordered."

2. Exceptions.
   2.1. The Manager may choose to abandon the order addition process at any time.

Includes: N/A
Notes/Issues: N/A

3.3.7. Manager Use Case - Order Log - Edit Order Status [MUC - OL - EOS]

Figure 11. Diagram for Use Case MUC - OL - EOS

Author/s: Cyril Andre Durango, Ivor Louisetyne Canque
Purpose: Allows the manager to edit orders stored in the order log database

Requirements Traceability:

Priority: High Priority. It allows immediate tracking of newly-arrived items

Preconditions: The Manager must be in the Order Log Screen, and a certain instance of an order has its status marked as “ordered”.
Postconditions: The status of the order has changed to “arrived”.

Actors: Manager
Extends: MUC - OL - VOL

Flow of Events:

1. Basic Flow
   1.1. The status column of the order log is a dropdown box.
   1.2. The Manager clicks the dropdown box and changes the status of the order to “arrived” once the stocks have arrived.
   1.3. A pop-up notification appears and warns the Manager to check if the items have arrived.
   1.4. The Manager proceeds with the order, and the status of the order is changed to “arrived.”
   1.5. The “date of arrival” box of the order instance will be filled on the exact date the order status was changed.
   1.6. The items in an order will immediately be accounted for in the software

2. Alternative Flow
   2.1. In step 4, the Manager can choose to discontinue changing the status of the order. This will prompt the system to go back to the Order Log Screen with its original status set.
   2.2. When such a case where a certain item in the inventory is full, it stores the excess products in a separate inventory.

3. Exceptions. The Manager may or may not opt to change the status of the order.

Includes: N/A
Notes/Issues: N/A

3.3.8. Manager Use Case - Order Log - View Excess Inventory [MUC - OL - VEI]

Figure 12. Diagram for Use Case MUC - OL - VEI

Author/s:
Purpose: Allows the manager to view excess inventory stored in the product and order database

Requirements Traceability:

Priority: High Priority. Often executed, isn't required by other use case

Preconditions: The Manager must be in the Order Log Screen
Postconditions: The excess inventory is accessed.

Actors: Manager
Extends: MUC - OL - VOL

Flow of Events:

1. Basic Flow
   1.1. The Manager clicks a button entitled "See Excess Inventory"
   1.2. The excess inventory is displayed, containing all the excess products and items brought by an influx of orders

2. Exceptions. The Manager may choose to or not to access the Excess Inventory.

Includes: N/A
Notes/Issues: N/A

3.3.9. Manager Use Case - Inventory Analysis Report - Pop-up Report [MUC - IAR - PuR]

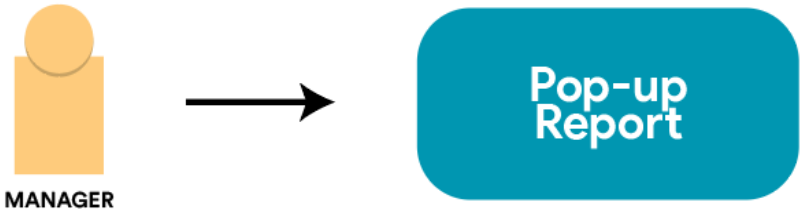

Figure 13. Diagram for Use Case MUC - IAR - PuR

Author/s:
Purpose: Informs the manager of the analysis report collected by the system at login

Requirements Traceability:

Priority: High Priority. Always executed, isn't required by other use case

Preconditions: Successful Authentication
Postconditions: N/A

Actors: Manager
Extends: MUC - OL - VOL
MUC - PD - VD
MUC - IAR - VAR

Flow of Events:

1. Basic Flow
   1.1. Once the Manager is granted access, they are immediately directed to the product database screen.
   1.2. Before they are to access the product database, they will be greeted with a pop-up report of the analysis report.
   1.3. The Manager choose to view the analysis report, and thus will be directed to the Inventory Analysis and Reporting screen.

2. Alternative Flow
   2.1. The manager can close the pop-up anytime and will be redirected to the product database screen

Includes: N/A
Notes/Issues: N/A

3.3.10. Manager Use Case - Inventory Analysis Report - View Analysis Report [MUC - IAR - VAR]

Figure 14. Diagram for Use Case MUC - IAR - VAR

Author/s:
Purpose: Informs the manager of the analysis report collected by the system at login

Requirements Traceability:

Priority: High Priority.

Preconditions: The Manager must press the “Analysis Report” tab on the upper part of the application or they pressed ‘proceed’ in the pop-up notification
Postconditions: N/A

Actors: Manager
Extends: MUC - PD - VD
MUC - IAR - PuR

Flow of Events:

1. Basic Flow
   1.1. The Manager clicks on the ‘proceed’ button on the pop-up notification
   1.2. The Manager is redirected to the Report Screen

2. Alternative Flow
   2.1. The Manager can also access the report screen by clicking on a tab entitled “Inventory Analysis and Report”

Includes: N/A
Notes/Issues: N/A

4 Other Non-functional Requirements

4.1 Performance Requirement

Performance requirements describe the software's ability to perform tasks like how fast it responds to certain requests, how large is its storage capacity, the integrity of data, etc. This should show how the software must excel in doing its tasks and ensure little to no errors when handling them.

4.1.1 Response Time

Our inventory software should be able to have a quick response time in handling out responses since they are essential to the users as they can keep track of stock counts and will enable them to have accurate data about their inventory. For example, using an inventory auditing system is where users can keep track of the amount of stocks like how much is left and how low the quantity of the stock is.

4.1.2 Storage Capacity

When we talk about inventory management, we should have a large storage capacity. Maintaining the right amount of inventory should at least ensure the user that there would be no sell-outs and reduce financial risks.

4.1.3 Data Integrity

The system should maintain the integrity of inventory data, ensuring that it is accurate, consistent, and up-to-date. This may involve implementing validation checks, error handling, and data validation mechanisms.

4.1.4 Usability

The system should have a user-friendly interface that makes it easy for users to perform inventory-related tasks efficiently. This includes intuitive navigation, informative feedback, and customizable settings.

4.1.5 Real-Time Data and Automation

The system should provide accurate, real-time data and automate business functions related to inventory management.

4.2 Safety and Security Requirements

The following list is the requirements that ensure all the data and information in the Inventory Software Management System (ISMS) are secured and safe. The system has built-in data backup and recovery, user authentication, and many more.

1. **Cybersecurity.** Ensures the inventory management system is safeguarded against unauthorized digital intrusions.

2. **Data Encryption.** Protects sensitive inventory data from being read or tampered with by unauthorized parties.
3. **Electronic Data Disposal.** Safely eliminates obsolete inventory records to prevent unauthorized access or recovery**.**
4. **User/Data Authentication.** Confirms the identity of users accessing the inventory system to secure data integrity.
5. **Backup and Recovery.** Maintains copies of inventory data to enable restoration in the event of data loss or system failure.

4.3 Software Quality Attributes

Software quality attributes ensure that our software runs smoothly and almost bug-free with high quality. However, it is not always guaranteed that a software that has little to no bugs will have a high quality. The following listed attributes will aid in the quality of Inventory Software Management System (ISMS) and show how reliable our software is.

4.3.1 Reliability

Inventory Software Management System (ISMS) should ensure the ability of the software to behave and function properly under a huge load of work.

4.3.2 Availability

When the software undergoes so much web traffic, there is a tendency that the software might crash and slow down because of its overwhelming traffic. We should know the percentage of the availability of the software and its tendency to crash due to web traffic. Usually, this is caused by a low bandwidth so providing a large bandwidth could make the software's availability even more reliable.

4.3.3. Recoverability

A crashed software should be able to recover quickly and efficiently for businesses or clients using the software to run back on track.

Appendix A – Data Dictionary

Inventory Report "inventoryCount" Data Dictionary

Field Name	Data Type	Data Format	Field Size	Description	Example
highestSerialNum	int	####	8	stores the highest serial number of a certain product	15
name	Text	XNNNNNNNNNN	20	The name of the product	Coca-Cola
price	float	####.##	10	the price of the product	500.00
quantity	Int	####	8	the current quantity of the products	30

Product Database "singleProduct" Data Dictionary

Field Name	Data Type	Data Format	Field Size	Description	Example
expirationDate	Timestamp	yyyy-MM-dd HH:mm:ss	26	Determines the expiration date of a certain product	June 8, 2024 at 8:00:00 AM
price	float	####.##	10	the price of each product	500.00
productName	Text	XNNNNNNNNNN	20	The name of the product	Coca-Cola
serialNum	int	###	8	The serial number of the product	2
status	String	STATUS_XNNNNNNN	16	Determines the status of the product in the	"STATUS_ON_DISPLAY"
supplierName	Text	XNNNNNNNNNN	20	Name of the Supplier	Coca Cola Co.
uniqueIdentifier	Text	XNNNNN	10	Made-up of the consonants of the product name.	CCCL

Order Log "orderedItems" Dictionary

Field Name	Data Type	Data Format	Field Size	Description	Example
date_arrived	Timestamp	yyyy-MM-dd HH:mm:ss	26	Determines the arrival date of the order	June 8, 2024 at 8:00:00 AM
date_request	Timestamp	yyyy-MM-dd HH:mm:ss	26	Determines the request date of the order	June 8, 2024 at 8:00:00 AM
prodExpiryDates	Timestamp Array	yyyy-MM-dd HH:mm:ss	26	An array of timestamps that show the expiry dates of products	June 8, 2024 at 8:00:00 AM
prodNames	String Array	[XNN, XNN, XNN]	n/a	an array of product names	[Sprite, Fanta, Coke]
prodPrices	Int Array	[###, ###, ###]	n/a	an array of product prices	[45, 50, 55]
prodQuantity	Int Array	[###, ###, ###]	n/a	an array of product quantity	[5, 2, 10]
status	String	STATUS_XNNNNNNN	16	Determines the status of the product in the	"STATUS_ON_DISPLAY"
supplierName	Text	XNNNNNNNNN	20	Name of the Supplier	Coca Cola Co.

Appendix B - Group Log

MINUTES OF THE MEETING

Activity Name	:	SRS Document Content Discussion
Date	:	March 2, 2024 - 9:00 PM
Speaker/s	:	Cyril Andre Durango
Venue	:	Google Meet
Meeting Duration	:	50 minutes
Attendees	:	Besinga, Piolo Pascual Canque, Ivor Louisetyne Ruiz, Jehu Lyndon
Absentees	:	N/A

Agenda:

1. Language/s to be used for the software
2. Dissemination of tasks for the SRS Document
3. Important Files to be Submitted

Agenda 1 [Languages to be used for the software]

For the software application, it was decided that the languages used for the software system are HTML, CSS, Java, Javascript, PHP, and MySql—HTML and CSS for the front-end design of the system and MySql for the back-end for the database of the inventory system

Agenda 2 [Dissemination of tasks for the SRS Document]

The team decided that every section of the SRS document has some people tasked to make it. But for the convenience of all, the team also decided that other people can help with the section that they are not assigned to have faster completion of the document

Agenda 3 [Important Files to be Submitted]

The team is reminded of all the important files to be submitted before the deadline.

Summary:

1. The team decided that these languages are used for the software system: HTML, CSS, Java, Javascript, PHP, MySql.
2. Every member has some assigned task for the SRS document.
3. Submit all the important file/s on or before the deadline
    a. SRS Document

Note: All important file/s must be submitted in pdf format. The deadline is on March 8, 2024.

Suggestions/Open Forum

1. Mr. Canque suggested changing the title of the software system.

2. The team is confused whether the system is a website or a standalone, but Mr. Durango remembered and is sticking to the proposals of the clients stating the application to be standalone. The application of JavaScript still remains.